

Tutorial 8 – Solutions

Q7. Explain the operation of MOS Capacitor with P- type semiconductor and sketch the energy band diagram for each mode operation.

- a) Accumulation mode
- b) Depletion mode
- c) Strong Inversion mode

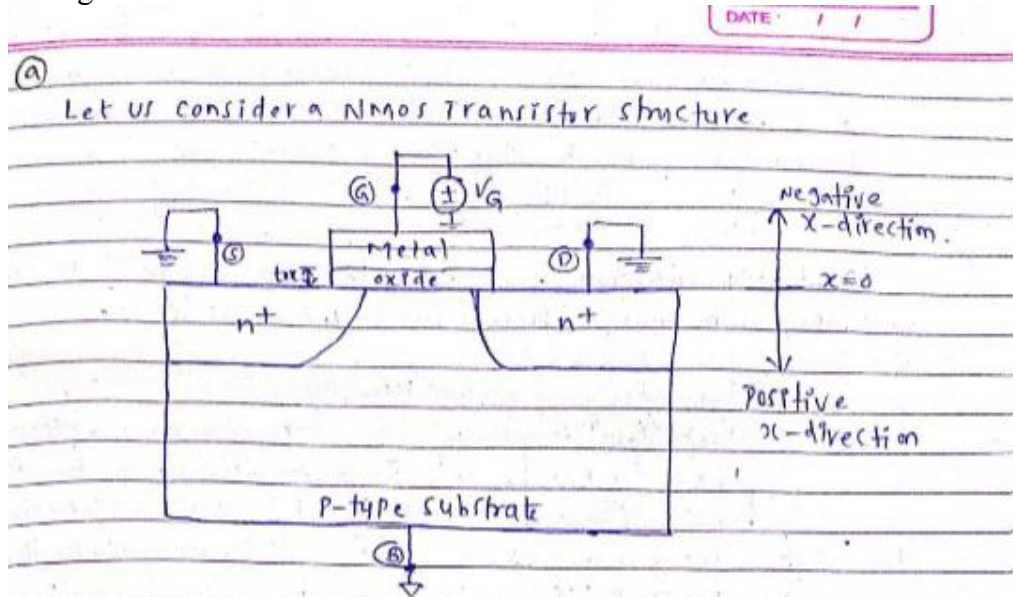


Fig NMos Transistor cross-sectional view.

Accumulation mode

→ Here, I am applying Negative gate voltage across the Mos device. This gate is generated Negative charges on the metal layer. Due to this Negative charges electric flux (D) is generated From ~~top to bottom~~ (gate to substrate) to top.

Indirectly, Electric field is created across the Mos device From substrate to gate (Electric field is negative direction). Due to Electric field, holes are accumulated (or) going towards the Si-SiO₂ interface (or) Surface.

Generally, When we apply gate voltage across the Mos device it will drop across the oxide and silicon. Now, we can write

$$V_G = V_{ox} + V_{si}$$

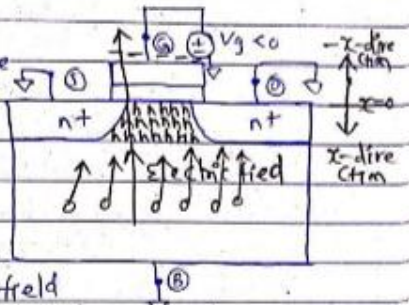


Fig NMos Transistor

$$V_G = V_{ox} + V_{si}$$

$V_G = E_{ox} t_{ox} + V_{si}$ (1) V_{ox} = voltage drop across the oxide layer at the Si-SiO₂ interface V_{si} = voltage drop across the silicon.

Electric flux is coming in oxide = Electric flux is going in silicon. We know that

$$E = \frac{V}{L}$$

NOTE: My assumption is SiO₂ on oxide layer is ideal. It means there is no charges present inside the oxide layer.

$$D_{ox} = D_{si}$$

Here,

$$(E_{ox})(\epsilon_{ox}) = (\epsilon_{si})(E_{si})$$

ϵ_{ox} = permittivity of oxide material

ϵ_{si} = permittivity of silicon material

E_{ox} = Electric field in the oxide ($V/\mu m$)

E_{si} = Electric field in the silicon ($V/\mu m$)

$$E_{ox} = \frac{\epsilon_{si} E_{si}}{\epsilon_{ox}}$$

Substitute E_{ox} into eq (1)

$$V_G = E_{ox} t_{ox} + V_{si}$$

$$V_G = t_{ox} \cdot \frac{\epsilon_{si} E_{si}}{\epsilon_{ox}} + V_{si}$$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

⇒ oxide capacitance per unit area

$$V_G = \frac{\epsilon_{si} E_{si}}{C_{ox}} + V_{si}$$

Form Gauss' law

$$V_G = \frac{1}{C_{ox}} (\epsilon_{si} E_{si}) + V_{si}$$

$$\nabla \cdot D = \rho$$

$$\nabla \cdot \epsilon E = \rho \Rightarrow \nabla \cdot E = \frac{\rho}{\epsilon}$$

If 1D (unimaterial)

$$Q_{si} = \epsilon_{si} E_{si}$$

$$\frac{dE(x)}{dx} = \frac{\rho(x)}{\epsilon}$$

$$V_G = \frac{1}{C_{ox}} Q_{si} + V_{si}$$

$$V_G = \frac{Q_{si}}{C_{ox}} + V_{si} \rightarrow (2)$$

$$dE(x) = \frac{1}{\epsilon} \rho(x) dx$$

Integrating on both sides

$$E(x) = \frac{1}{\epsilon} \int \rho(x) dx$$

$$E_{si}(x) = \frac{1}{\epsilon_{si}} \int \rho(x) dx$$

$$\epsilon_{si}(x) \epsilon_{si} = \int \rho(x) dx$$

↓
This will represent total charge in the silicon.

$$\text{If } V_G = V_{ox} + V_{si}$$

$$V_{si}(x=0) = \psi_s$$

$$V_G = \text{negative}$$

$$\psi_s \Rightarrow \text{is also negative}$$

$$\epsilon_{si} E_{si} = Q_{si}$$

$$V_G = \frac{Q_{Si}}{C_{ox}} + \psi_s$$

Q_{Si} = Total charge in silicon in Accumulation mode

$$Q_{Si} = \sqrt{2\epsilon_s \epsilon_0 N_A K T} \exp\left(-\frac{q\psi_s}{K T}\right)$$

$$V_G = \frac{1}{C_{ox}} \left[\sqrt{2\epsilon_s \epsilon_0 N_A K T} \exp\left(-\frac{q\psi_s}{K T}\right) + \psi_s \right]$$

Depletion mode

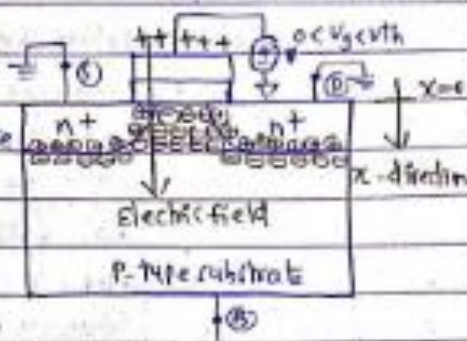
We know that

$$V_G = V_{ox} + \psi_s$$

→ Here, we are applying positive gate voltage across the device. This positive gate voltage will generate the electric flux in the top to bottom direction (gate to substrate).

Due to this electric flux indirectly electric field is also generated. ($D = \epsilon E$).

Now, Electric field is positive x-direction due to holes are moving towards the Bulk regions (or) depleting towards the Bulk region in the substrate. Here, holes are leaving behind the Negative Immobile Negative Ion.

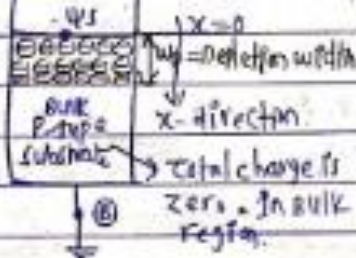


$$V_G > (0 < V_G < V_{th})$$

* Here, I am applying gate V_G varying from 0 to Threshold voltage.

$$V_G \uparrow \Rightarrow W_D \uparrow$$

$V_G \uparrow \Rightarrow$ More holes are depleted $\Rightarrow$ more negative Immobile ions are generated.



$$V(x > W_D) = 0 \quad \text{because of} \\ \epsilon(x > W_D) = 0 \quad \text{Bulk region.}$$

→ Due to this Immobile Negative Ions, voltage is generated across the oxide layer. The corresponding voltage is (ψ_s) or ψ_s .

ψ_s = surface potential @ voltage is at $x=0$ in silicon.

$$V_G = V_{ox} + \psi_s$$

$V_G \uparrow \Rightarrow \psi_s \uparrow$. Here surface potential increases as a positive. Now we have ~~two~~ n-p junctions in the nmos transistor. These n-p junctions becomes F.B (Forward bias). If any junction is forward, charges carrier [electrons] are injected from source to substrate as well as a drain to substrate. These electrons are experiencing a electric field in top to bottom but this electric field is weak due to this all the electrons are not Accumulating at the surface.

Now, we can say

Q_D = Depletion charge

W_D = depletion width

$$Q_D = q N_A W_D$$

$$W_D = \sqrt{\frac{2 \epsilon_s \epsilon_0 \psi_s}{q N_A}}$$

$$Q_D = \sqrt{2 q N_A \epsilon_s \epsilon_0 \psi_s}$$

Q_{inv} = Inversion charge

$$Q_{inv} = q n_s t_{inv}$$

$Q_D \gg Q_{inv} \Rightarrow$ Depletion mode.

$$V_G = \frac{Q_{si}}{C_{ox}} + \psi_s = \frac{Q_D + Q_{inv}}{C_{ox}} + \psi_s = \frac{Q_D}{C_{ox}} + \psi_s$$

$$V_G = \frac{1}{C_{ox}} \sqrt{2 q \epsilon_s \epsilon_0 N_A (\psi_s)} + \psi_s$$

Inversion mode

We know that

$$V_G = V_{ox} + \psi_s$$

$$V_G = \frac{Q_{si}}{C_{ox}} + \psi_s$$

$\rightarrow$ we can modeled, this

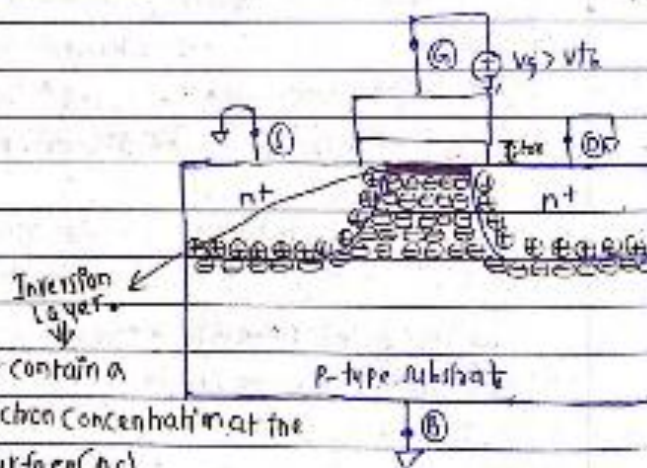
Nmos transistor

into two capacitor

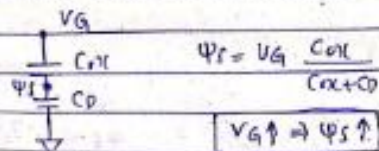
are in series.

It contain a

Electron concentration at the surface (n_s)



→ Here, I am increasing gate voltage more and more. Due to this surface potential is also increases.



→ Due to increasing a surface potential, both n⁺p Junctions are more forward bias. Now, more electrons are injected from source to substrate and drain to substrate. Now, more and more electrons are distributed in the depletion region due to strong electric field will exist b/w gate to substrate. These electrons are experiencing the force, electrons are accumulating towards the surface. At some point electron concentration at the surface is greater than substrate doping concentration. This is called a Inversion region.

$$Q_{inv} = -q n_s t_{inv} \quad t_{inv} = \text{Thickness of the Inversion layer.}$$

$V_G > V_{th} \Rightarrow$ strong inversion

$$Q_{inv} = -q n_s t_{inv}$$

$$Q_D = -q N_A w_D$$

$$Q_{inv} \gg Q_D$$

$$V_G = \frac{Q_{si}}{C_{ox}} + \psi_s = \frac{Q_{inv} + Q_D}{C_{ox}} + \psi_s$$

$$V_G = \frac{Q_{inv}}{C_{ox}} + \psi_s$$

$$V_G = \frac{Q_{inv}}{C_{ox}} + \psi_s = \frac{1}{C_{ox}} \sqrt{2 \epsilon_s q N_A} \sqrt{e^{\frac{q \psi_s}{kT}} - 1} + \psi_s$$

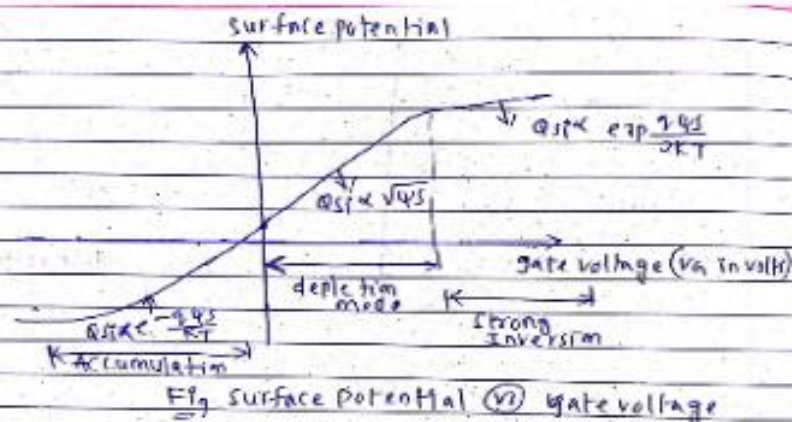
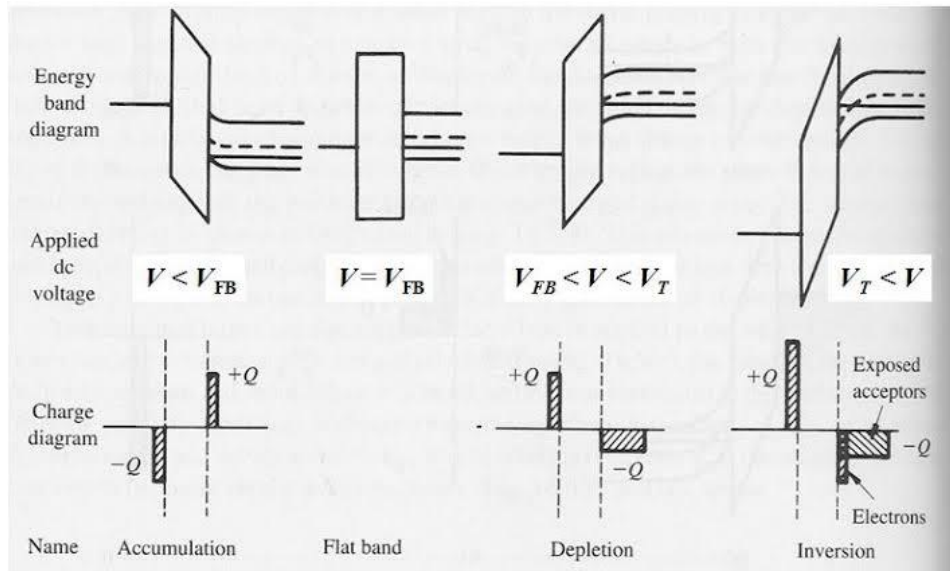
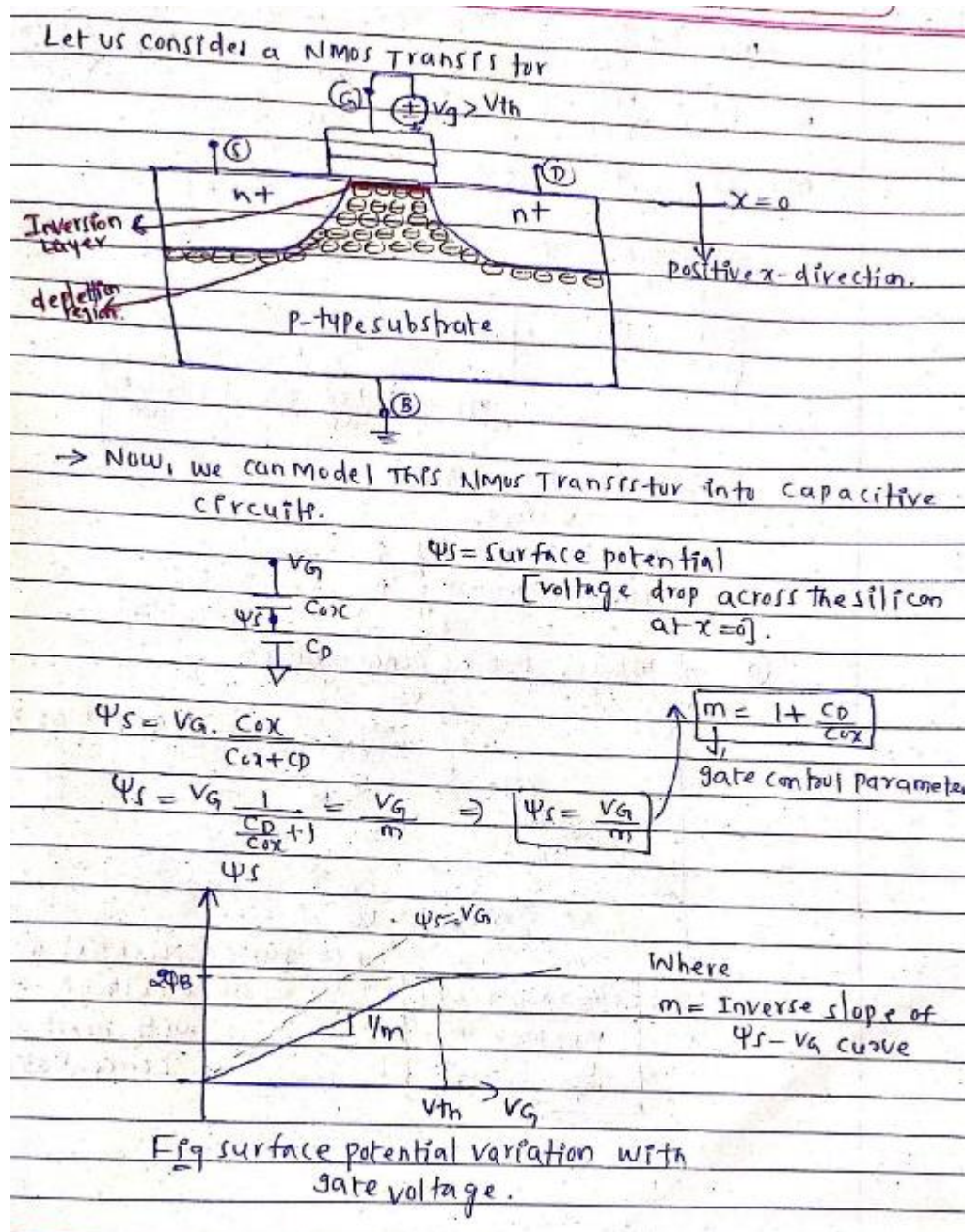


Fig. Surface Potential (ψ_s) vs gate voltage



Q8. For the MOSCAP with P-Type substrate.

a) Plot the graphs for surface potential versus gate voltage



1) Channel doping variation

$$m = 1 + \frac{C_D}{C_{ox}}$$

$$m = 1 + C_D \frac{t_{ox}}{\epsilon_{ox}}$$

$$m = 1 + \frac{\epsilon_{si}}{w_D} \cdot \frac{t_{ox}}{\epsilon_{ox}}$$

$$\frac{\epsilon_{si}}{\epsilon_{ox}} = \frac{(\epsilon_{r,si} \epsilon_0)}{(\epsilon_{r,ox} \epsilon_0)}$$

$\frac{\epsilon_{si}}{\epsilon_{ox}} = \frac{11.7}{4} \approx 3$

$C_D = \text{depletion capacitance}$

$$C_D = \frac{\epsilon_{si}}{w_D} \quad w_D = \sqrt{\frac{2\epsilon_{si}\psi_s}{qN_A}}$$

$$C_D = \frac{\epsilon_{si}}{\sqrt{2\epsilon_{si}\psi_s}} \sqrt{2N_A}$$

$C_D = \sqrt{\frac{\epsilon_{si} 2 N_A}{2\psi_s}}$

$m = 1 + \frac{t_{ox}}{w_D} 3 = 1 + 3 \frac{t_{ox}}{w_D}$

$$m = 1 + 3 \frac{t_{ox}}{\sqrt{2\epsilon_{si}\psi_s}} \sqrt{2N_A}$$

* $t_{ox} \downarrow \Rightarrow m \downarrow \Rightarrow \frac{1}{m} \uparrow$

* $N_A \downarrow \Rightarrow m \downarrow \Rightarrow \frac{1}{m} \uparrow$

(a) Substrate doping concentration

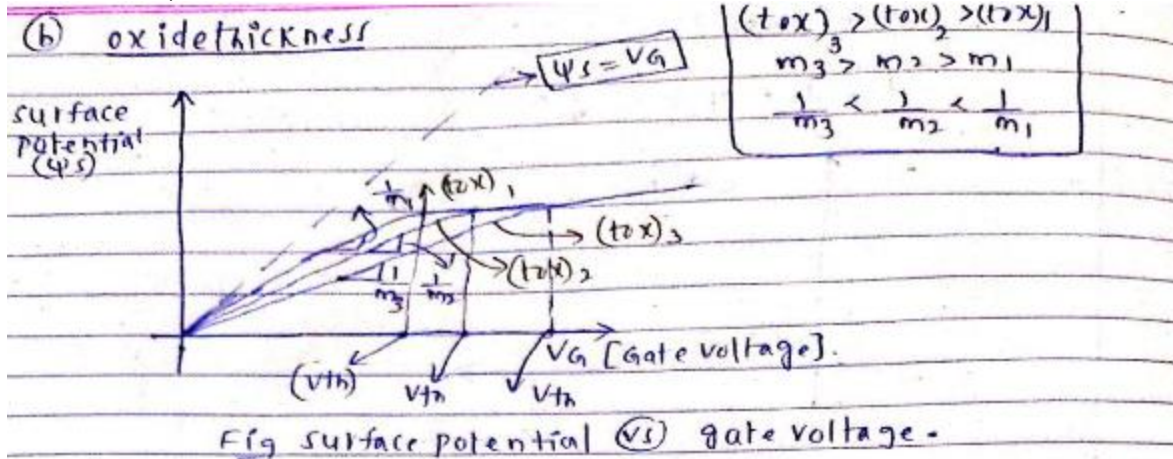
$N_{A3} > N_{A2} > N_{A1}$

$N_{A3} > N_{A2} > N_{A1}$
 $m_3 > m_2 > m_1$
 $\frac{1}{m_3} < \frac{1}{m_2} < \frac{1}{m_1}$

Fig surface potential (ψ_s)

gate voltage variation with substrate doping concentration.

2) Oxide thickness variations



b) Why surface potential saturates after threshold voltage? Give the proper reason.

Why (ψ_s) is saturates in Strong Inversion?

→ In Strong Inversion, gate voltage is ~~very~~ high compare to the threshold voltage. ~~very to this~~

$V_g \uparrow \Rightarrow \psi_s \uparrow$

Increasing surface potential, Both n+p Junctions are forward bias. More and more electrons are injected from source ~~and~~ as well as drain into the substrate. It means very high thick sheet of electrons are created at the surface. Now, electric field is exist from top to bottom but due to this sheet of electrons electric field is not crossing. If without electric field, depletion region will not increases due to this surface potential (ψ_s) is almost (or) approximately saturates.

Q9. For the MOSCAP with P-Type substrate.

- Plot the graphs for silicon charge versus surface potential
- Plot the graphs for silicon charge versus gate voltage

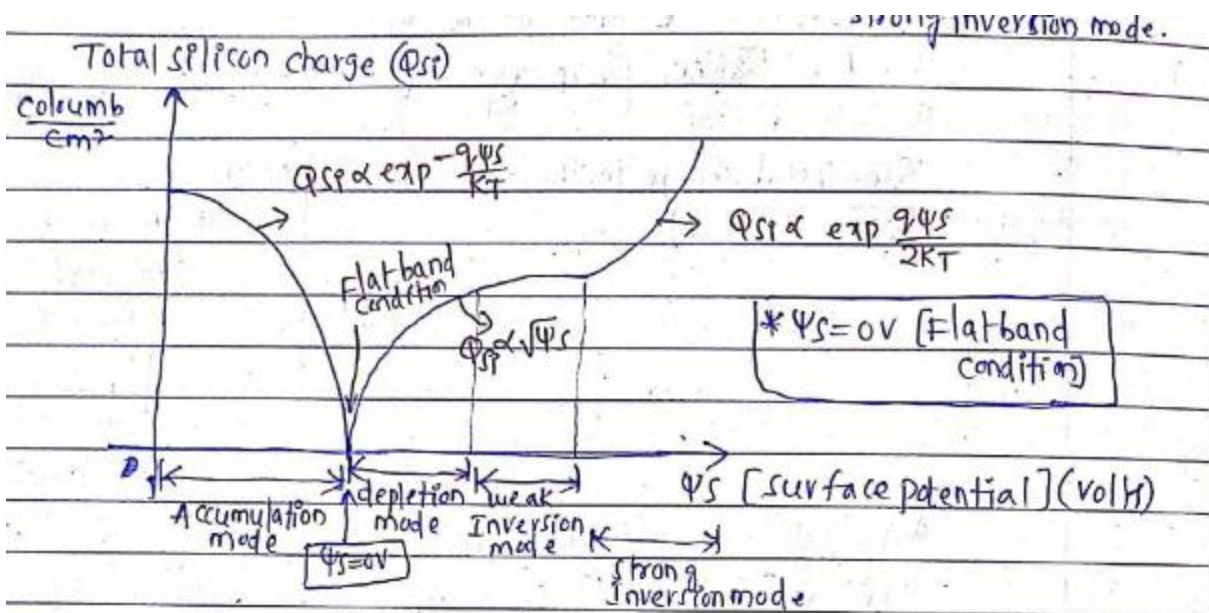


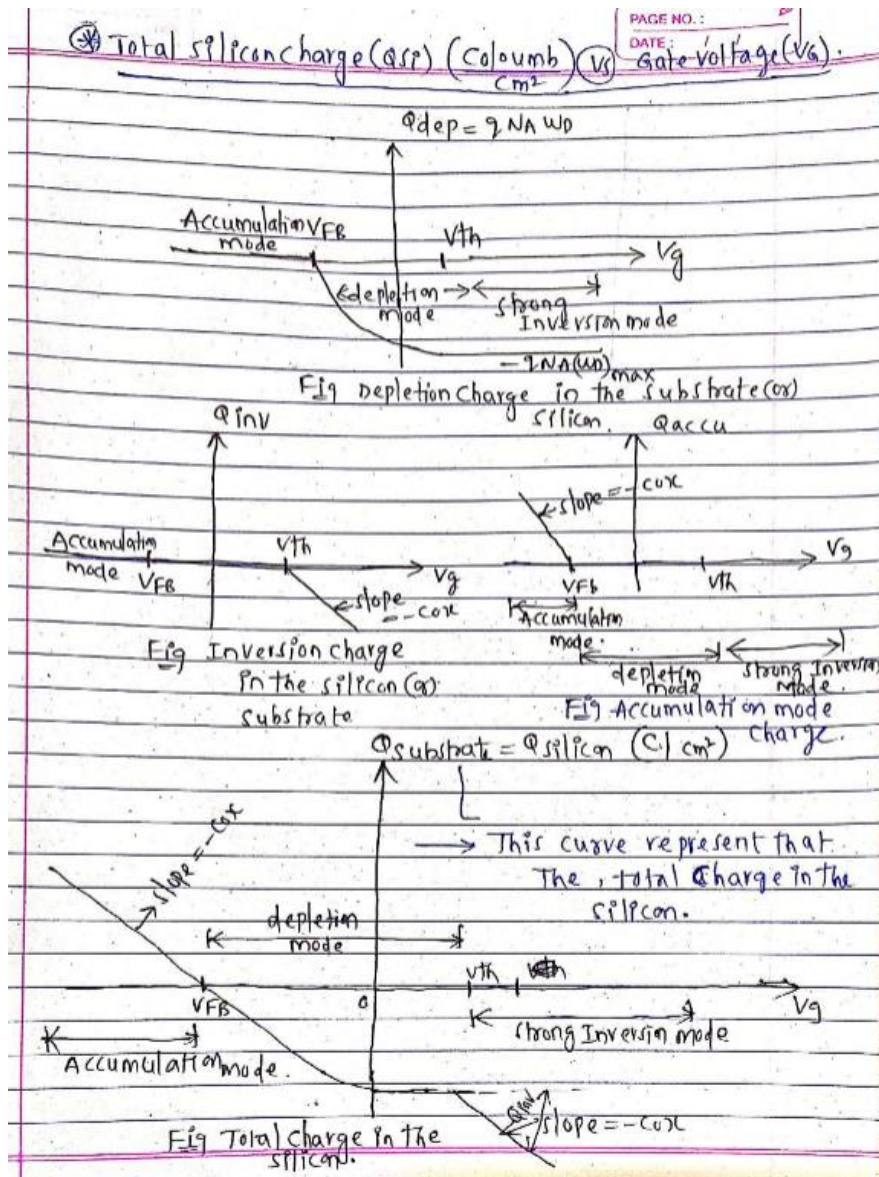
Fig Total silicon charge variation with surface.

* $Q_{si} \propto \exp\left(-\frac{q\psi_s}{K_T}\right) \Rightarrow$ Accumulation mode

* $Q_{si} = 0 \Rightarrow$ Flatband condition

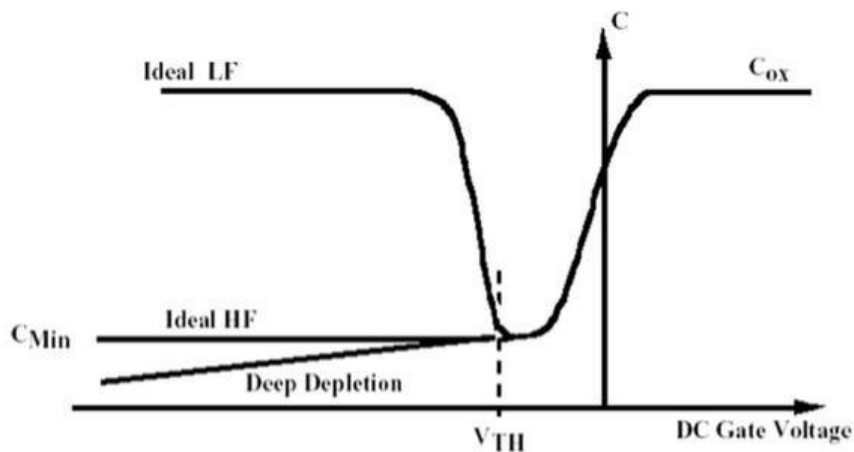
* $Q_{si} \propto \sqrt{\psi_s} \Rightarrow$ Depletion and weak Inversion mode

* $Q_{si} \propto \exp\left(\frac{q\psi_s}{2K_T}\right) \Rightarrow$ strong Inversion mode.



Q10. For the MOSCAP with P-Type substrate.

- a) Plot the C-V characteristics of the HFCV, LFCV and Deep Depletion CV, Explain the behavior and give the proper reason.



High frequency Explanation

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If Input signal Frequency is very high (KH2, MH2)
 $\downarrow$
 It changes the voltage very fast
 $\downarrow$
 I am Not giving enough time to the minority carrier to be generated in the bulk to move in near the surface.
 So, Total Capacitance of the system is equal series of oxide capacitance and depletion capacitance.

Low Frequency

In this Mode

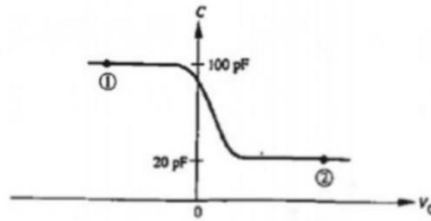
Total capacitance (C') = (C_{ox}) series with the C_D

$$\frac{1}{C'} = \frac{1}{C_{ox}} + \frac{1}{C_D}$$

$$\frac{1}{C'} = \frac{C_D + C_{ox}}{C_{ox} C_D}$$

$$C' = \frac{C_{ox} C_D}{C_{ox} + C_D} \Rightarrow C' = \frac{C_D}{1 + \frac{C_D}{C_{ox}}}$$

Q11. The CV characteristics exhibited by a MOSCAP is displayed in the figure below:



- Are the semiconductor components of MOSCAP n-type or p-type?
- If the area of MOS-C is $3 \times 10^{-3} \text{ cm}^2$ what is the thickness of oxide?
- Determine the depletion width.

Solution 5: (a) P-type

(b) and (c):

$$C_{\max} = C_O = \frac{K_O \epsilon_0 A_G}{x_0}$$

$$x_0 = \frac{K_O \epsilon_0 A_G}{C_{\max}} = \frac{(3.9)(8.85 \times 10^{-14})(3 \times 10^{-3})}{(10^{-10})} = 0.104 \mu\text{m}$$

delta depletion formation,

$$C = \frac{C_O}{1 + \frac{K_O W_T}{K_S x_0}} \quad \text{inv } (\omega \rightarrow \infty) \quad (16.)$$

Thus

$$W_T = \frac{K_S x_0}{K_O} \left(\frac{C_O}{C} - 1 \right) = \frac{(11.8)(0.104)}{(3.9)} \left(\frac{100}{20} - 1 \right) = 1.26 \mu\text{m}$$

Q12. A MOS capacitor consists of a p-type silicon substrate and an n^+ poly-silicon gate. The oxide thickness is $22 \text{ \AA}$ (2.2 nm), the substrate doping concentration is $N_A = 3 \times 10^{17} \text{ cm}^{-3}$, and the gate doping concentration is $N_D = 1 \times 10^{20} \text{ cm}^{-3}$. The intrinsic carrier concentration is $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$. The relative permittivity of silicon and silicon dioxide are **11.7** and **3.9**, respectively. The oxide layer contains 2×10^{10} positive ions per cm^2 (i.e., $Q_{ox} = 2 \times 10^{10} \times q$, where $q = 1.6 \times 10^{-19} \text{ C}$).

Using the given data, calculate the following parameters for the MOS capacitor:

- The oxide capacitance per unit area.
- The maximum depletion width in silicon.
- The maximum electric field in silicon and in the oxide.
- The flat-band voltage.
- The threshold voltage.

Answer :- (1) Oxide Capacitance per unit area

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-13}}{2.2 \times 10^{-7}} = 1.57 \times 10^{-6} \text{ F/cm}^2$$

$$C_{ox} = 1.57 \mu\text{F/cm}^2$$

(2) Maximum depletion width in silicon

At threshold:

$$\psi_s = 2\phi_F$$

where

$$\phi_F = \phi_t \ln \left(\frac{N_A}{n_i} \right) = 0.026 \ln \left(\frac{3 \times 10^{17}}{1.45 \times 10^{10}} \right) = 0.026 \times 17.85 = 0.464 \text{ V}$$

Hence,

$$2\phi_F = 0.928 \text{ V}$$

Now,

$$\begin{aligned} W_{max} &= \sqrt{\frac{2\epsilon_{Si} \times 2\phi_F}{qN_A}} \\ W_{max} &= \sqrt{\frac{2 \times 1.035 \times 10^{-12} \times 0.928}{1.6 \times 10^{-19} \times 3 \times 10^{17}}} \\ &= \sqrt{\frac{1.92 \times 10^{-12}}{4.8 \times 10^{-2}}} = \sqrt{4.0 \times 10^{-11}} = 6.32 \times 10^{-6} \text{ cm} \end{aligned}$$

$$W_{max} = 6.3 \times 10^{-6} \text{ cm} = 63 \text{ nm}$$

(3) Maximum Electric Field

At threshold, electric field in silicon surface:

$$\begin{aligned} E_{Si,max} &= \frac{qN_A W_{max}}{\epsilon_{Si}} \\ &= \frac{1.6 \times 10^{-19} \times 3 \times 10^{17} \times 6.32 \times 10^{-6}}{1.035 \times 10^{-12}} = 2.93 \times 10^5 \text{ V/cm} \end{aligned}$$

Electric field in oxide (using dielectric ratio):

$$E_{ox} = \frac{\epsilon_{Si}}{\epsilon_{ox}} E_{Si} = \frac{11.7}{3.9} \times 2.93 \times 10^5 = 8.78 \times 10^5 \text{ V/cm}$$

$$E_{Si,max} = 2.93 \times 10^5 \text{ V/cm}, E_{ox} = 8.8 \times 10^5 \text{ V/cm}$$

(4) Flat-band Voltage

$$V_{FB} = \phi_{MS} - \frac{Q_{ox}}{C_{ox}}$$

Work-function difference:

$$\phi_{MS} = \phi_M - \phi_S$$

For n⁺ poly-Si, $\phi_M \approx \chi_{Si} + (E_g/2) - \phi_F(n^+) \approx 4.05 + 0.56 - (-0.5) \approx 5.11$ eV.

For p-type Si, $\phi_S = \chi_{Si} + (E_g/2) + \phi_F = 4.05 + 0.56 + 0.464 = 5.07$ eV

$$\Rightarrow \phi_{MS} = 5.11 - 5.07 = 0.04 \text{ V}$$

Now,

$$\frac{Q_{ox}}{C_{ox}} = \frac{3.2 \times 10^{-9}}{1.57 \times 10^{-6}} = 2.04 \times 10^{-3} \text{ V}$$

$$V_{FB} = 0.04 - 0.002 \approx 0.038 \text{ V}$$

(5) Threshold Voltage

$$\begin{aligned} V_T &= V_{FB} + 2\phi_F + \frac{\sqrt{4\varepsilon_{Si}qN_A\phi_F}}{C_{ox}} \\ &= 0.038 + 0.928 + \frac{\sqrt{4(1.035 \times 10^{-12})(1.6 \times 10^{-19})(3 \times 10^{17})(0.464)}}{1.57 \times 10^{-6}} \end{aligned}$$

The numerator term:

$$\begin{aligned} 4\varepsilon q N_A \phi_F &= 9.2 \times 10^{-14} \Rightarrow \sqrt{9.2 \times 10^{-14}} = 3.03 \times 10^{-7} \\ \frac{3.03 \times 10^{-7}}{1.57 \times 10^{-6}} &= 0.193 \end{aligned}$$

$$V_T = 0.038 + 0.928 + 0.193 = 1.16 \text{ V}$$

Q14. Calculate the threshold voltage of a silicon nMOS capacitor with a substrate doping $N_a = 10^{17} \text{ cm}^{-3}$, a 20 nm thick oxide ($\varepsilon_{ox} = 3.9 \varepsilon_0$) and an aluminum gate ($\Phi_M = 4.1 \text{ V}$). Assume there is no fixed charge in the oxide or at the oxide-silicon interface.

Solution: The threshold voltage equals:

$$\begin{aligned} V_T &= V_{FB} + 2\phi_F + \frac{\sqrt{4\varepsilon_s q N_a \phi_F}}{C_{ox}} \\ &= -0.93 + 2 \times 0.42 \\ &\quad + \frac{\sqrt{4 \times 11.9 \times 8.85 \times 10^{-14} \times 1.6 \times 10^{-19} \times 10^{17} \times 0.42}}{3.9 \times 8.85 \times 10^{-14} / 20 \times 10^{-7}} \\ &= -0.09 \text{ V} \end{aligned}$$

Where the flatband voltage was already calculated in the previous question. The threshold voltage voltages for nMOS and pMOS capacitors with an aluminum or a poly-silicon gate are listed in the table below.

	Aluminum	p ⁺ poly	n ⁺ poly
nMOS	-0.09 V	0.98 V	-0.14 V
pMOS	-0.93 V	0.14 V	-0.98 V

Q15. Calculate the oxide capacitance, the flatband capacitance and the high frequency capacitance in inversion of a silicon nMOS capacitor with a substrate doping $N_a = 10^{17} \text{ cm}^{-3}$, a 20 nm thick oxide ($\epsilon_{ox} = 3.9 \epsilon_0$) and an aluminum gate ($\Phi_M = 4.1 \text{ V}$).

Solution: The oxide capacitance equals:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.85 \times 10^{-14}}{2 \times 10^{-6}} = 173 \text{ nF/cm}^2$$

The flatband capacitance equals:

$$C_{FB} = \frac{1}{\frac{1}{C_{ox}} + \frac{L_D}{\epsilon_s}} = \frac{1}{\frac{1}{173 \times 10^{-9}} + \frac{1.3 \times 10^{-6}}{11.9 \times 8.85 \times 10^{-14}}} = 142 \text{ nF/cm}^2$$

where the Debye length is obtained from:

$$L_D = \sqrt{\frac{\epsilon_s V_t}{q N_a}} = \sqrt{\frac{11.9 \times 8.85 \times 10^{-14} \times 0.0259}{1.6 \times 10^{-19} \times 10^{17}}} = 13 \text{ nm}$$

The high frequency capacitance in inversion equals:

$$C_{HF,inv} = \frac{1}{\frac{1}{C_{ox}} + \frac{x_{d,T}}{\epsilon_s}} = \frac{1}{\frac{1}{173 \times 10^{-9}} + \frac{1.05 \times 10^{-5}}{11.9 \times 8.85 \times 10^{-14}}} = 63 \text{ nF/cm}^2$$

and the depletion layer width at threshold equals:

$$x_{d,T} = \sqrt{\frac{2\epsilon_s(2\phi_F)}{qN_a}}$$

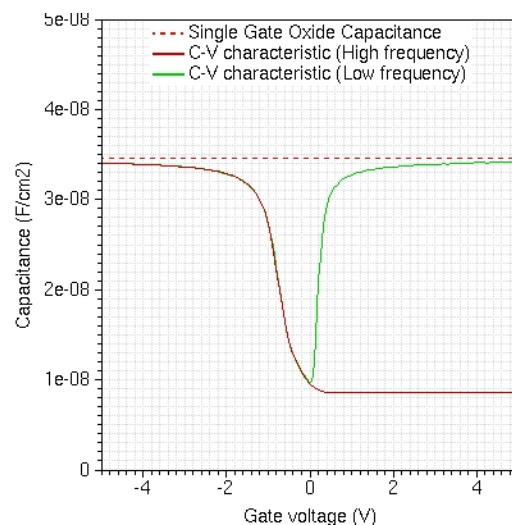
$$= \sqrt{\frac{2 \times 11.9 \times 8.85 \times 10^{-14} \times 2 \times 0.419}{1.6 \times 10^{-19} \times 10^{17}}} = 105 \text{ nm}$$

Please note that the bulk potential(Φ_F) was already calculated above in Q13.

Q17. For a MOSCAP with SiO₂ as oxide and Si as semiconductor, a CV measurement is shown.

Answer following:

- Calculate the Oxide thickness.
- What is the type of the semiconductor?
- Calculate the doping of the semiconductor.
- Calculate the Threshold voltage V_{th} (Assume work function of metal and semiconductor are equal i.e. $\phi_m = \phi_{Si}$)
- Draw the band-diagram in equilibrium and inversion & mark the values. (Assume $\phi_m = \phi_{Si}$)
- Find the midgap voltage and midgap capacitance
(Midgap voltage is the gate voltage at which the doped Si behaves as if intrinsic exactly at the oxide/silicon interface)



if $\phi_m \neq \phi_{Si}$

- a. Calculate the value of Flatband capacitance, Flatband Voltage using the same CV measurements.

Calculate the actual threshold voltage.

Solution:

$$a) \quad C_{ox} = \frac{\epsilon_{SiO_2} A}{t_{ox}} ; \quad \frac{C_{ox}}{A} = 3.4 \times 10^{-8} \text{ F/cm}^2$$

$$\Rightarrow t_{ox} = \frac{8.85 \times 10^{-14} \times 4}{3.4 \times 10^{-8}} = 104 \times 10^{-7} \text{ cm}$$

$t_{ox} = 104 \text{ nm}$

b) p-type as inversion can be seen on positive polarity

c) Doping can be found from High Freq CV in inversion region where C_{min} can be seen. Here, C_{ox} & C_{dep} capacitance are in series. C_{dep} is result of max depletion width at $\psi_s = 2\phi_B$.

$$\frac{1}{C_{min}} = \frac{1}{C_{ox}} + \frac{1}{C_{dep}}$$

$$C_{min} \approx 8 \times 10^{-9} \text{ F/cm}^2$$

$$C_{ox} = 3.4 \times 10^{-8} \text{ F/cm}^2$$

$$\Rightarrow C_{dep} = 1.04 \times 10^{-8} = \frac{\epsilon_{Si}}{W_{max}} = \frac{8.85 \times 10^{-14} \times 12}{W_{max}}$$

$$\Rightarrow W_{max} = \sqrt{\frac{2 \epsilon_{Si} (2\phi_B)}{q N_A}} = 10^{-4} \text{ cm} \quad \text{--- (1)}$$

$$\Rightarrow \frac{\phi_B}{N_A} = 5.1 \times 10^{-16}$$

$$\phi_B = \frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right) = 5.1 \times 10^{-16} \text{ NA}$$

given $n_i = 1.1 \times 10^{10} / \text{cm}^3$

$$\Rightarrow N_A = 5.09 \times 10^{13} \ln(N_A/n_i) \quad \text{--- (2)}$$

Solve iteratively.

Assume some value of N_A .

$$\text{say } N_A = 10^{17} / \text{cm}^3$$

Put in RHS of (2)

$$\Rightarrow \text{New } N_A = 8.16 \times 10^{14} / \text{cm}^3$$

Put this new N_A again in RHS of (2)

$$\Rightarrow \text{New } N_A = 5.7 \times 10^{14} / \text{cm}^3$$

Put again in RHS of (2)

$$\Rightarrow \text{new } N_A = 5.5 \times 10^{14} / \text{cm}^3$$

$$\text{So, } N_A \text{ is } \approx \underline{\underline{5.5 \times 10^{14} / \text{cm}^3}}$$

$$d) V_T = V_{FB} + V_{ox} + \psi_s$$

$$V_{FB} = 0 \quad \text{as given } \phi_m = \phi_{si}$$

$$\Rightarrow V_{ox} = \frac{-Q_{dep}}{C_{ox}} = \frac{qNAW_{max}}{C_{ox}}$$

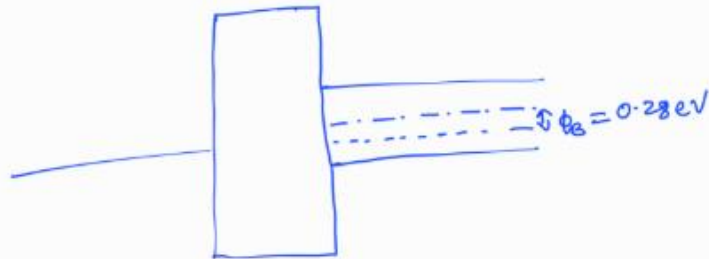
(mark) { from (1)
 $W_{max} = 10^{-4} \text{ cm}$
 $C_{ox} = 3.4 \times 10^{-8} \text{ F/cm}^2$ from part (a)
 $N_A \approx 5.5 \times 10^{14} \text{ /cm}^3$ from part (c)

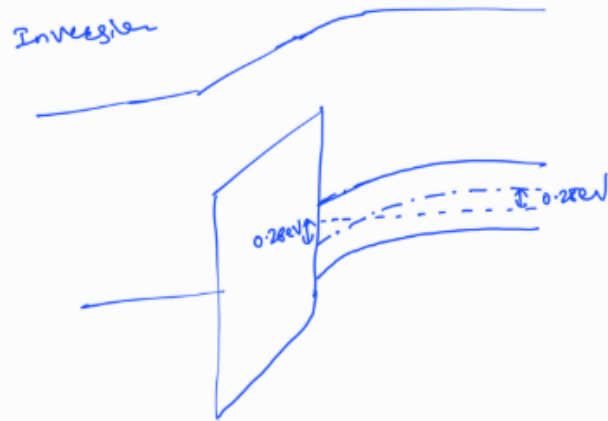
$$\Rightarrow V_{ox} = 0.26 \text{ V}$$

(mark) { $\psi_s = 2\phi_B$ (at inversion)
 $= 2V_T \ln(N_A/n_i)$;
 $\psi_s = 0.56 \text{ V}$

$$\Rightarrow \boxed{V_{To} = 0.26 + 0.56 = 0.82 \text{ V}} \quad - (3)$$

e) Equilibrium of p-si





f) midgap voltage V_{MG} (@ $\psi_s = \phi_B$)

$$V_{GS} = V_{MG} = \underset{\downarrow 0}{V_{FB}} + \psi_s + V_{ox}$$

$$\Rightarrow \psi_s = \phi_B = 0.28 \text{ eV} \quad \left. \vphantom{\psi_s = \phi_B} \right\} \text{(1 mark)}$$

$$V_{ox} = \frac{-Q_{dep}}{C_{ox}} = \frac{q N_A w}{C_{ox}}$$

$$\text{So, } w = \sqrt{\frac{2 \epsilon_{Si} \psi_s}{q N_A}} \quad \left. \vphantom{w = \sqrt{\frac{2 \epsilon_{Si} \psi_s}{q N_A}}} \right\} \text{(1 mark)}$$

$$\Rightarrow w = 8.2 \times 10^{-5} \text{ cm}$$

$$\Rightarrow V_{ox} = 0.21 \text{ V}$$

$$\Rightarrow V_{MG0} = 0.21 + 0.28 = 0.49 \text{ V}$$

from graph @ $V = 0.49 \text{ V}$, $\left. \vphantom{V = 0.49 \text{ V}} \right\} \text{(1 mark)}$

$$C = C_{MG0} \approx C_{min}$$

if $\phi_m \neq \phi_{Si}$

Flatband situation occurs when the bands are flat.

When bands are flat the charge screening is given by Debye length.

$$\frac{1}{C_{FB}} = \frac{1}{C_{ox}} + \frac{1}{C_{Debye}}$$

$$C_{Debye} = \frac{\epsilon_{Si}}{L_{Debye}} = \frac{\epsilon_{Si}}{\sqrt{\frac{\epsilon_{Si} kT}{q^2 N_A}}}$$

$$C_{Debye} = \sqrt{\frac{q^2 N_A \epsilon_{Si}}{kT}}$$

$$C_{Debye} = 6 \times 10^{-8} \text{ F/cm}^2$$

$$\Rightarrow C_{FB} = 2.17 \times 10^{-8} \text{ F/cm}^2$$

from graph @ $C_{FB} = 2.17 \times 10^{-8} \text{ F/cm}^2$
 (1 mark) $\Rightarrow V = V_{FB} \approx -0.8 \text{ V}$

h)

$$V_T = V_{FB} + V_{T0}$$

Actual: $V_T = -0.8 + 0.82 = 0.02 \text{ V}$

(The moscap is already in inversion due to V_{FB})

Actual: $V_{MG} = V_{FB} + V_{MG0}$

$$V_{MG} = -0.8 + 0.49 = -0.31 \text{ V}$$

from graph @ $V_{MG} = -0.31 \text{ V}$

$$\Rightarrow C_{MG} \approx 1.2 \times 10^{-8} \text{ F/cm}^2$$

Q18. Derive the sub-threshold slope for MOSFET. What could be an ideal subthreshold slope for the MOSFET? Can this ideal SS be achieved? What are the limitations and challenges?

Answer: The sub-threshold slope (SS) represents how much gate voltage is required to change the drain current by one order of magnitude (one decade) when the MOSFET is

"off" ($V_{GS} < V_{TH}$). It is defined as:

$$SS = \frac{kT}{q} \ln(10) \left(1 + \frac{C_{dep}}{C_{ox}} \right) \quad \text{"mV/dec"}$$

$\underbrace{\hspace{10em}}_{60 \text{ mV/dec (ideal)}} \quad \downarrow \quad 0 \text{ (ideally)}$

The ideal value is achieved when the depletion capacitance (C_{dep}) is negligible compared to the oxide capacitance (C_{ox}), or when the gate has perfect control over the channel.

- **Ideal SS:** $\sim 60 \text{ mV/decade}$ at room temperature.

Can this ideal SS be achieved? No, because C_{dep} is always > 0 in bulk MOSFET.

$$SS = \frac{kT}{q} \ln(10) \left(1 + \frac{C_{dep}}{C_{ox}} \right) \quad \text{"mV/dec"}$$

$\underbrace{\hspace{10em}}_{60 \text{ mV/dec (ideal)}} \quad \downarrow \quad 0 \text{ (ideally)}$

I_{off} (Subthreshold)

- 1) $C_{dep} \rightarrow 0$; $\frac{\epsilon_{Si}}{w_{dep}} \rightarrow \alpha \frac{1}{\sqrt{V_{th}}} = \infty$
- 2) $C_{ox} \rightarrow \infty$ (limitation)

$\left(\frac{\epsilon_{ox}}{t_{ox}} \right) : t_{ox} \rightarrow 0 \neq 0$ (limitation)

